

FreshMart Sales Analytics

Phase 2: Exploratory Data Analysis Report

Objective:

With the data cleaned and validated in Phase 1, this phase explores the FreshMart sales dataset to answer business questions about sales trends, product performance, store performance, customer segments, and returns. Each question below uses pandas for the underlying calculations and a chart to show the pattern, with the insight written in plain language so it can be read on its own, without opening the notebook.

Question 1:

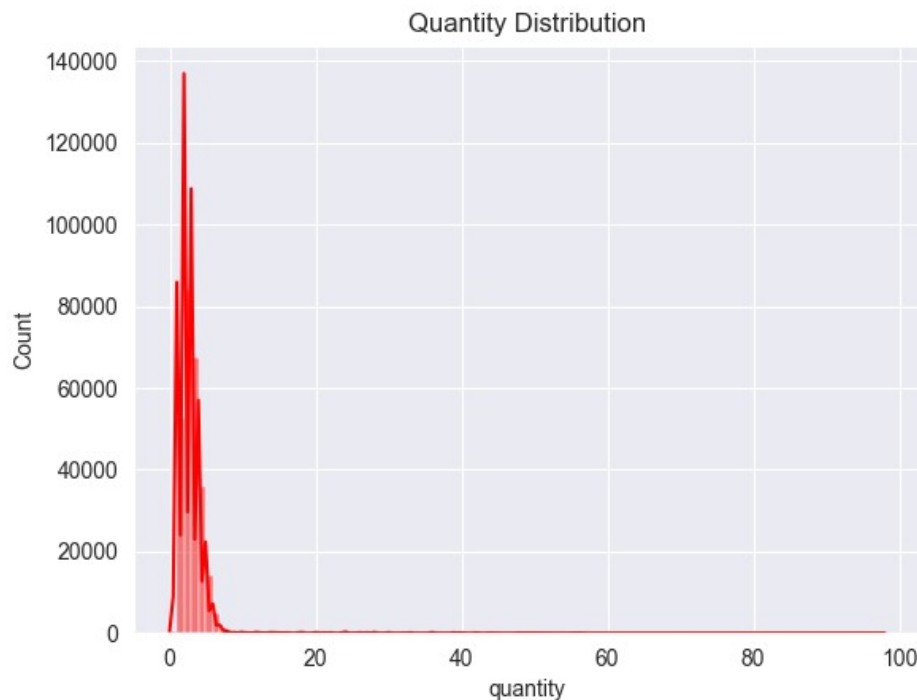
How does transaction volume change over time across 1997-1998, and are there clear monthly or seasonal demand patterns?



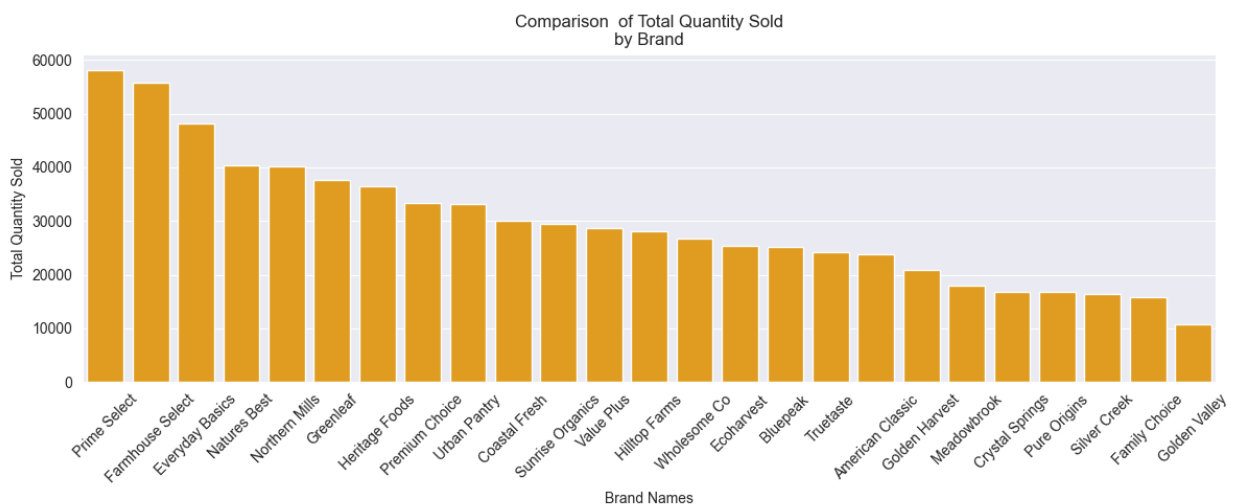
- Transaction volume follows a broadly similar monthly pattern across both years. Volume declines in **February**, increases sharply in **March**, and generally rises through the summer months before falling in September. It then recovers in October and increases sharply in November, with December recording a further increase. **November and December are consistently the strongest months, while February consistently records one of the lowest volumes.**
- The repeated pattern across both years suggests some seasonality in transaction activity, although a longer time period would be needed to confirm a strong seasonal trend with more confidence.

Question 2:

Which product brands contribute the most to sales quantity, and how concentrated is overall demand among the top performers?



Before comparing brands, I checked the underlying quantity distribution for outliers. It is highly right-skewed (**skewness = 11.34**), with most transactions involving **1-5 units**. A small number of transactions have much higher quantities, but these occur across a range of recurring values rather than as isolated spikes, and appear to represent legitimate bulk purchases rather than data errors. No quantity outliers were removed for this reason.



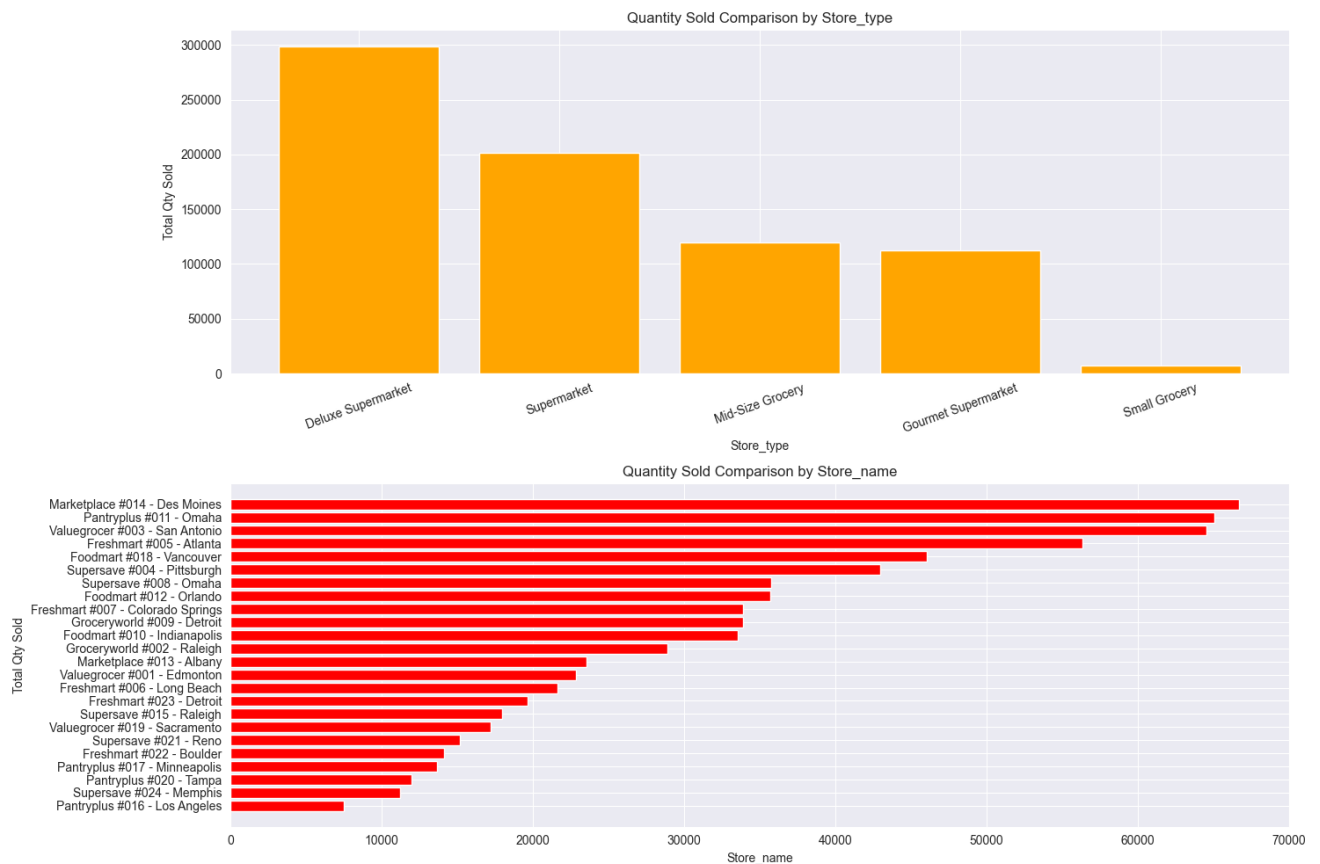
- **Prime Select** has the highest total quantity sold, followed by **Farmhouse Select** and **Everyday Basics**. Prime Select sold approximately **58K units**, while Farmhouse Select and Everyday Basics sold around **56K and 48K units**, respectively. **Demand is moderately concentrated among the top-performing brands,**

but not dominated by only a few — several brands in the middle of the ranking also contribute substantial volume.

- **Key business insight:** FreshMart should prioritise inventory availability for the leading brands, particularly Prime Select and Farmhouse Select, while continuing to monitor mid-performing brands for opportunities to grow their contribution.

Question 3:

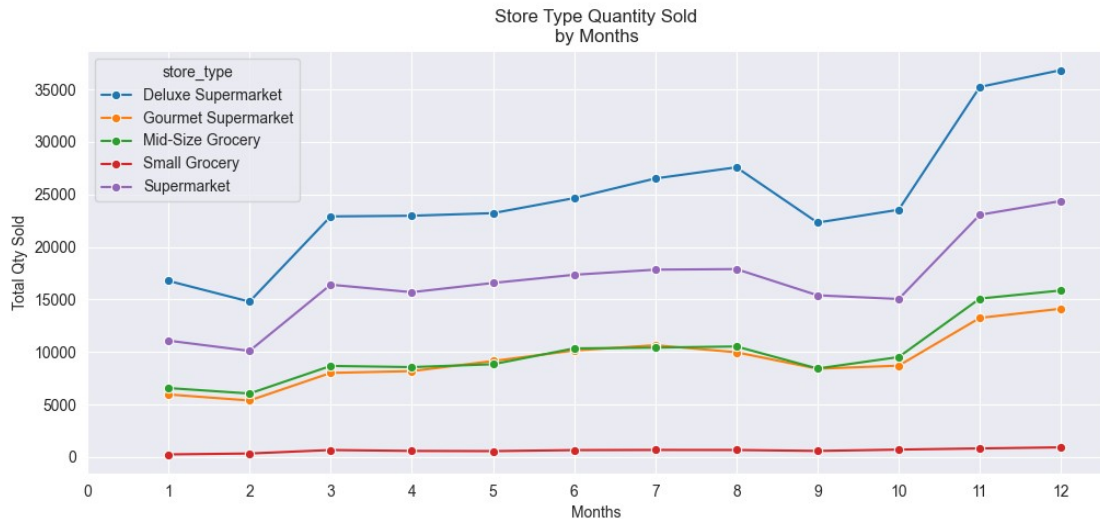
How does sales performance differ across store types and individual stores, and which stores consistently outperform or underperform?



- **Deluxe Supermarkets** sold the highest total quantity, followed by **Supermarkets**. Mid-Size Grocery and Gourmet Supermarket recorded fairly similar totals, while Small Grocery stores had the lowest total quantity sold among the store types.
- At the individual store level, **Marketplace #014 (Des Moines)** recorded the highest quantity sold, followed by **Pantryplus #011 (Omaha)** and **Valuegrocer #003 (San Antonio)**. The top-performing stores account for a substantial share of total quantity, but sales are spread across multiple stores rather than concentrated

in just one or two, with several mid-ranked stores also contributing meaningful volume.

- **Important note:** This analysis is based on quantity sold, not revenue — a store with higher quantity sold does not necessarily generate higher revenue, since revenue also depends on product prices and product mix.

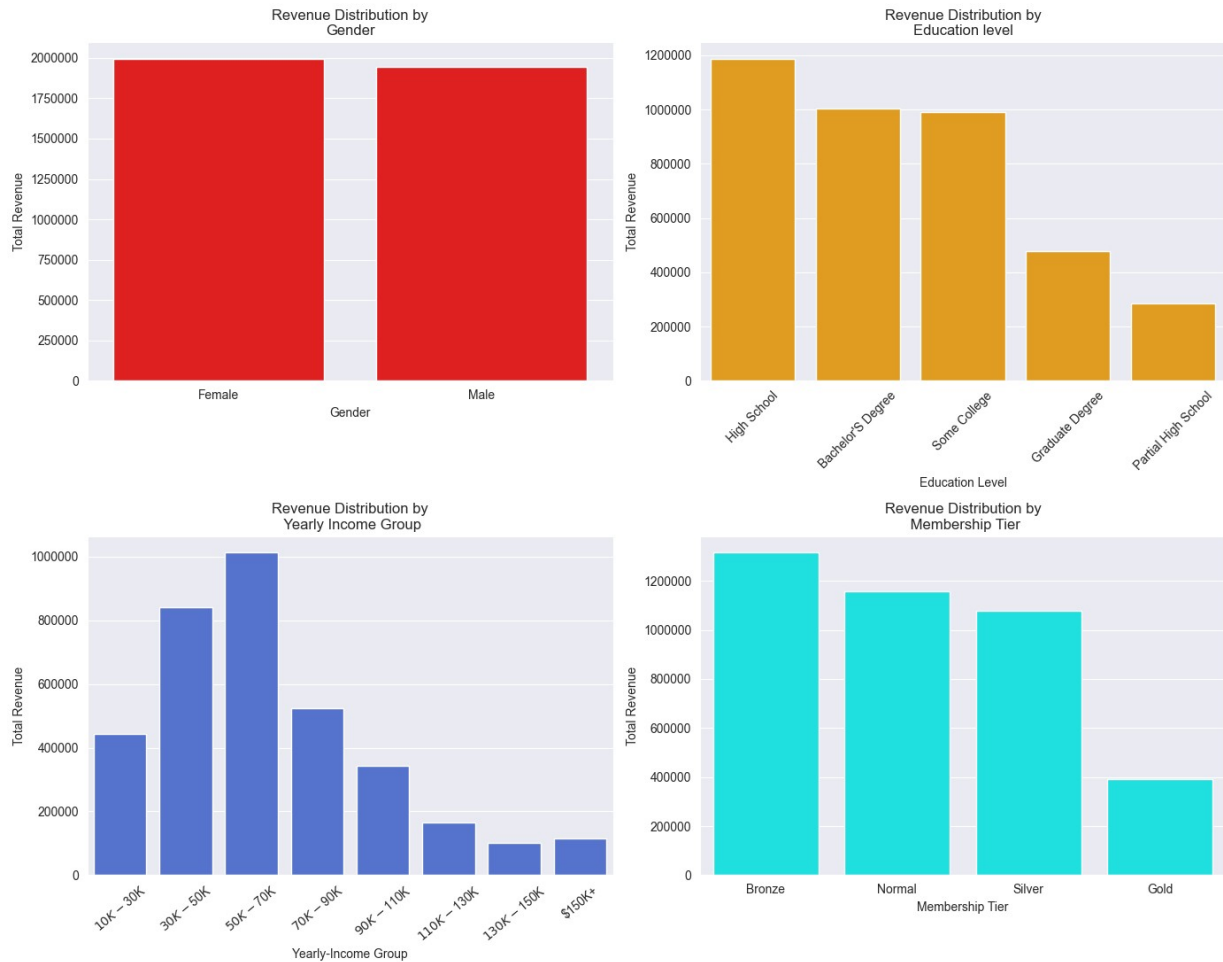


- **Deluxe Supermarket** is consistently the strongest performer, recording the highest quantity sold in every month. **Supermarket** is consistently second-best, staying well above the remaining store types throughout the year. **Mid-Size Grocery** generally outperforms **Gourmet Supermarket**, especially in the later months, though the two stay close. **Small Grocery** is consistently the weakest performer by a large margin, remaining far below all other store types throughout the year.

Question 4:

Which customer segments generate the most revenue, and how does revenue contribution differ across income, education, gender, and membership card categories?

Revenue Distribution across different customer segment



Gender: Female and male customers generated approximately similar revenue, with female customers contributing slightly more.

Education: Customers with a High School education level generated the highest revenue, followed by Bachelor's Degree and Some College. Partial High School customers generated the lowest revenue.

Income: The \$50K-\$70K income group generated the highest revenue, followed by \$30K-\$50K. Customers earning \$150K+ generated the lowest revenue among the income groups.

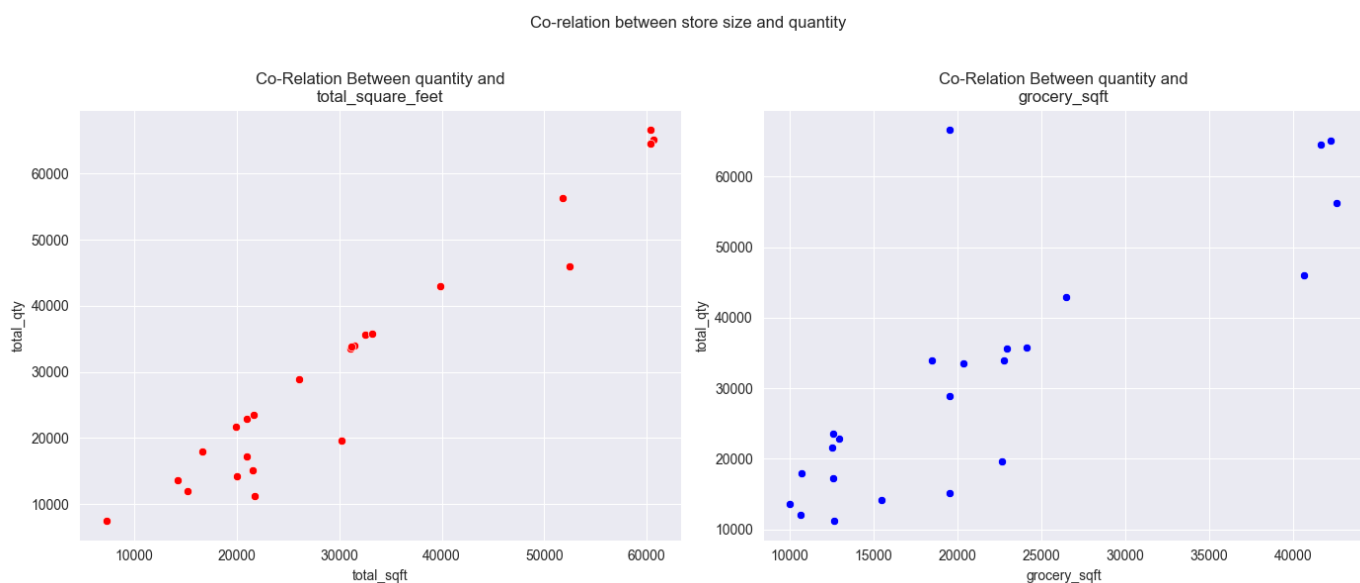
Membership: Bronze members generated the highest revenue, followed by Normal and Silver. Gold members generated the lowest revenue.

Important consideration: these totals reflect overall revenue per segment but do not account for the number of customers within each segment. A segment generating less total revenue may simply have fewer customers rather than lower spending per customer.

- **Business insight:** the next step is to compare customer count and revenue per customer across these segments. For lower-revenue segments such as Partial High School and Graduate Degree customers, this would clarify whether the gap is driven by a smaller customer base or genuinely lower spending — and if customer counts are comparable, their product/category purchasing patterns are worth investigating to understand the difference and identify opportunities to better serve them.

Question 5:

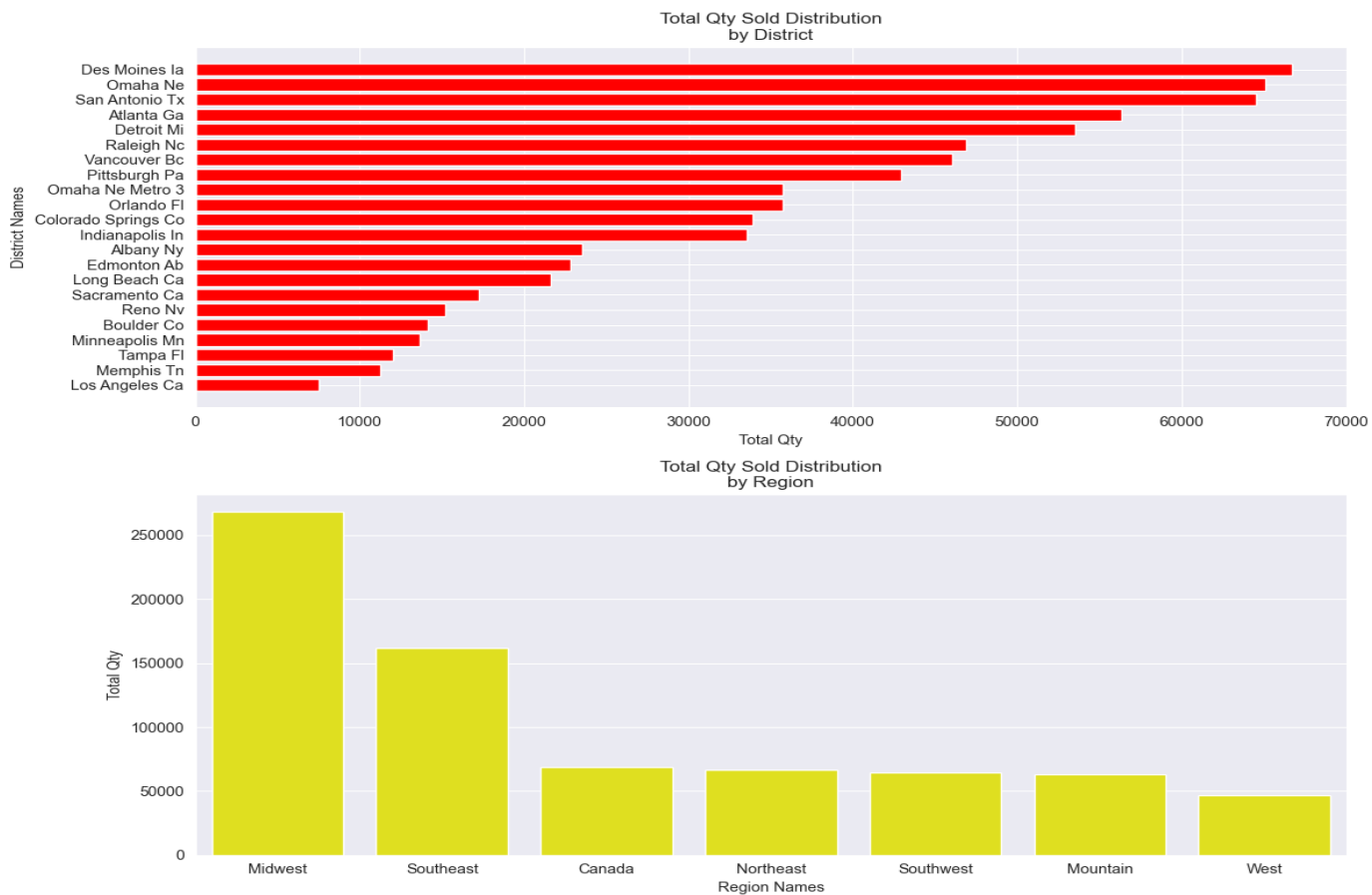
Do larger stores actually generate higher transaction volumes, or is store size a weak predictor of operational performance?



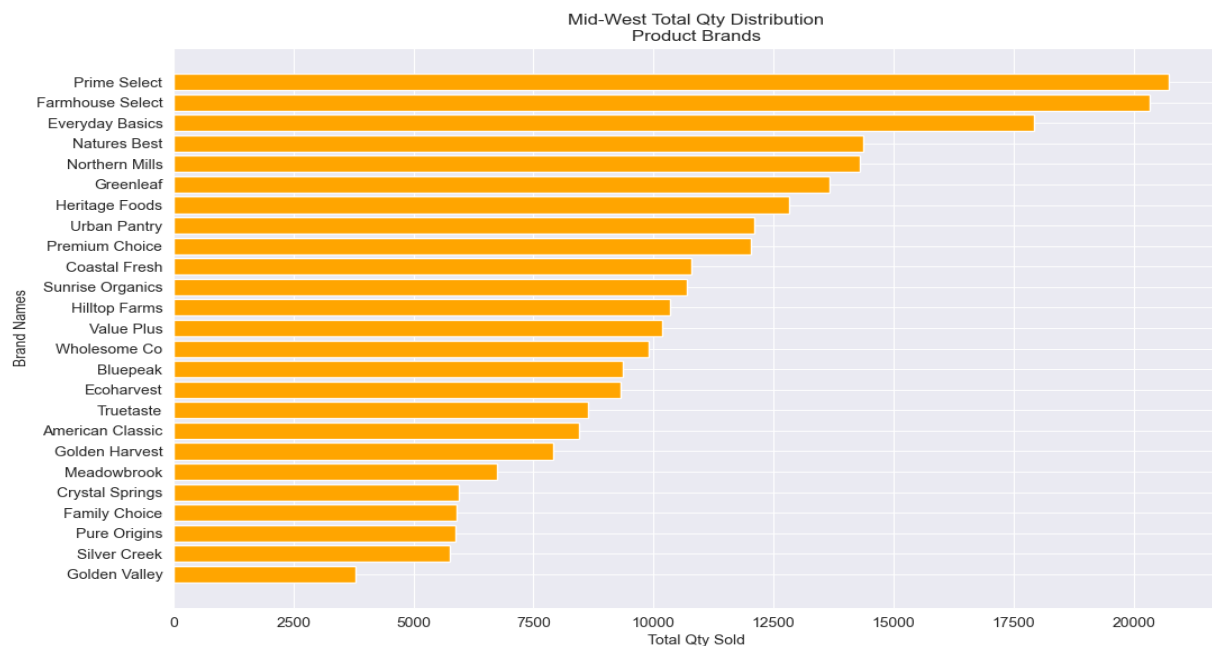
- The correlation between total store size (total_sqft) and total quantity sold is **0.9695** — a **very strong positive relationship**. The correlation between grocery floor area (grocery_sqft) and total quantity sold is **0.8165** — **also a strong positive relationship**, though noticeably weaker than total store size. Both scatter plots show a clear upward trend, with larger stores generally generating higher transaction quantities.
- **Overall, store size is a strong predictor of transaction volume, and the relationship is particularly strong for total square footage** — suggesting total store size is more closely associated with transaction volume than grocery area specifically.

Question 6:

Are there particular regions or districts where customer purchasing patterns differ substantially from the company-wide pattern?



- The **Midwest region** has the highest quantity sold by a substantial margin, followed by the **Southeast region** — with a significant gap between the two. Most of the remaining regions have relatively similar sales volumes, while the West region records slightly lower sales by comparison.
- At the district level, **Des Moines, IA** has the highest quantity sold, followed by **Omaha, NE** and **San Antonio, TX**. These three districts are the top contributors to overall volume, though the remaining districts also make meaningful contributions.

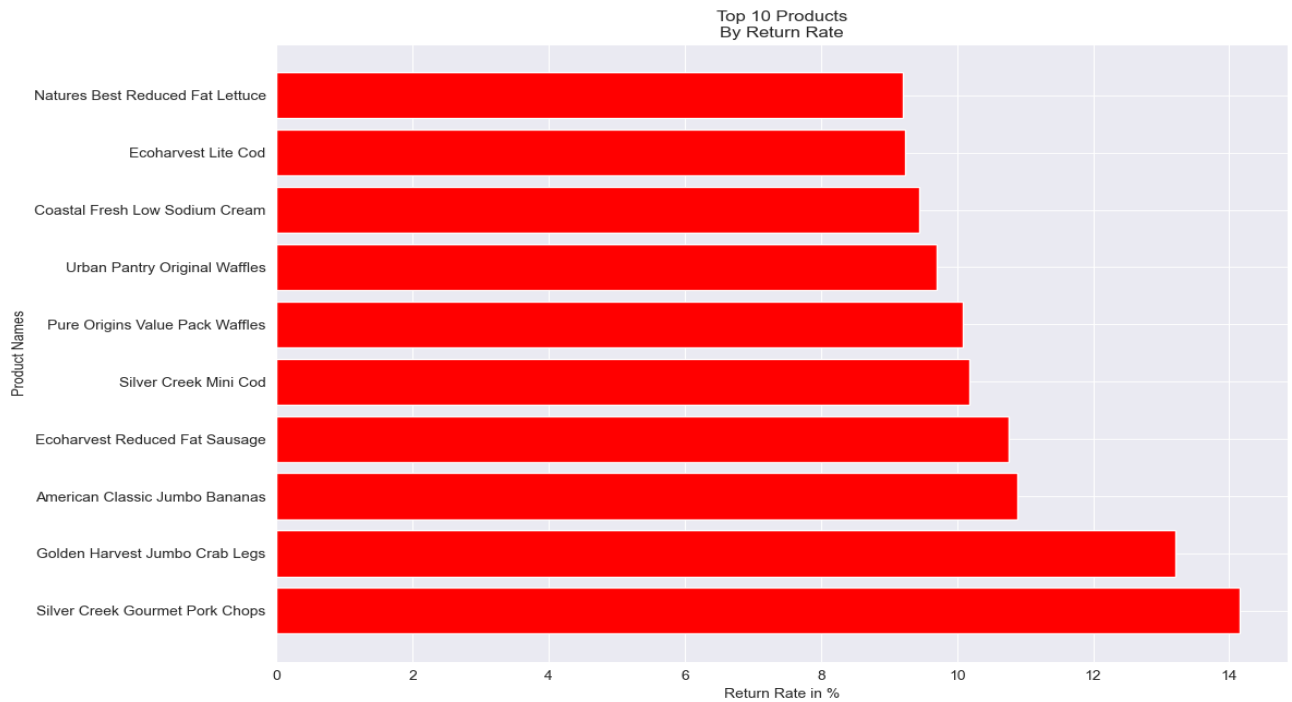


- To understand what's driving Midwest demand, I broke it down by product brand. **Prime Select and Farmhouse Select** have the highest quantities sold within the region, followed by **Everyday Basics and Natures Best** — but most other brands also contribute meaningful volume, with a gradual decline across the ranking rather than a sharp drop-off. **This suggests the Midwest's high demand is not dependent on a small number of specific brands and is relatively well distributed across the product portfolio.**

Question 7:

Which products and stores have the highest return activity relative to their transaction volume?

- Silver Creek Gourmet Pork Chops** has the highest return rate at approximately **14.1%**, followed by **Golden Harvest Jumbo Crab Legs** at around **13.2%**. Return rates then drop noticeably — **American Classic Jumbo Bananas and Ecoharvest Reduced Fat Sausage** sit at approximately **10.8% and 10.7%**, respectively, and the remaining products in the top 10 fall below 10%, with **Natures Best Reduced Fat Lettuce the lowest of the group at approximately 9.0%.**
- The clear gap between the top two products and the rest suggests Silver Creek Gourmet Pork Chops and Golden Harvest Jumbo Crab Legs are worth investigating further to understand what's driving their comparatively high return rates.



- **Supersave #015 (Raleigh)** has the highest store-level return rate at approximately **3.2%**, followed by **Pantryplus #017 (Minneapolis)** at around **3.1%** and **Freshmart #022 (Boulder)** at approximately **3.0%**. The remaining stores in the top 10 fall between roughly **2.8% and 3.0%**, so overall the group is fairly close together, with Raleigh and Minneapolis standing out only slightly.
- These stores are worth a closer look to see whether their higher return rates are tied to specific products, product categories, or customer purchasing patterns.

